

Optical Flicker Generator - User Manual

Purpose

This device generates controlled LED light output for optical flicker experiments. You can operate it from a browser on your local network, or via one of several APIs.

Quick Start

1. Connect the hardware:
 - Connect Ethernet to your LAN.
 - Power the device.
 2. Wait for startup to complete (display shows mode and settings).
 3. Open the control page:
 - Use the IP shown on the device display, or
 - Try `http://flicker-xxxxxx.local/` (where xxxxxx is the device suffix)
 4. Use the **Control Page** to set mode and output parameters.
 5. If needed, open **Configuration** from the control page to change network settings.
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Accessing the Web Interface

- Main control page: GET `/`
- Configuration page: GET `/config`

If `*.local` names do not resolve on your computer, use the device IP address shown on the display.

Using the Control Page

The control page is for direct use and testing.

Mode Selection

You can select one of four modes:

- **Off**: output disabled
- **Constant**: continuous light at selected intensity
- **Flicker**: square-wave modulation using frequency + duty cycle + intensity
- **Sinus**: sinusoidal modulation using frequency + intensity

Parameters

- **Frequency (Hz):** 1-500
- **Duty cycle (%):** 10-90
- **Intensity (%):** 0-100

Click **Apply** to send the selected mode and parameter values to the device.

Typical Workflow

1. Choose mode (flicker, sinus, constant, or off)
 2. Set frequency and intensity (and duty for flicker)
 3. Click **Apply**
 4. Verify output on the device and experiment setup
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Using the Configuration Page

Open /config from the control page when you need to change network or hardware-related settings.

Network Settings

- **Use DHCP** (checkbox)
 - Enabled: router assigns IP automatically
 - Disabled: device uses static values entered below
- **Static IP settings** (used when DHCP is disabled):
 - IP address
 - Gateway
 - Subnet mask

Connectivity Settings

- **Enable Telnet access (port 23)**
Disabled by default. Enable only on trusted networks.

Hardware/Display Settings

- **PWM carrier frequency (Hz):** 1,000-50,000
- **Display screensaver timeout (s):** 1-600

Click **Save** to store changes.

When Changes Take Effect

- Network-related changes are applied after a power cycle.

- Other saved settings are applied immediately.
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Factory Reset (Set Button)

- **Short press:** wakes display from screensaver.
 - **Press and hold for 10 seconds:** restores network and configuration defaults.
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Troubleshooting

Cannot open the control page

- Check Ethernet cable and LAN connection.
- Confirm the device has an IP address on the display.
- Try opening the page by IP instead of *.local hostname.

Wrong or no light output

- Confirm mode is not Off.
- Verify parameter ranges on the control page.
- Click **Apply** after any change.

Lost access after network changes

- Power-cycle the device.
- If static IP is misconfigured, perform factory reset and reconnect using DHCP defaults.